CS333 Midterm 3 Part (a)

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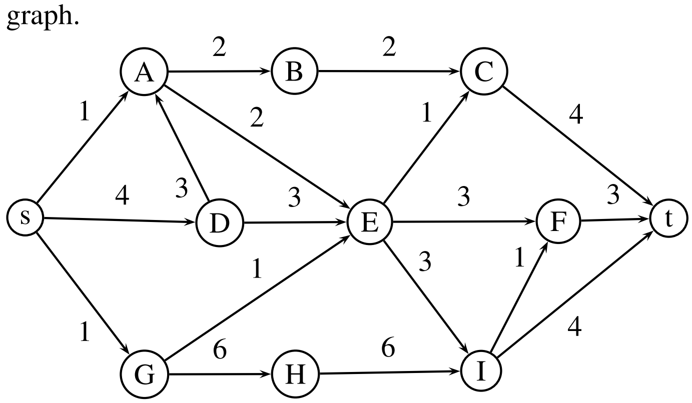
I. Multiple Choice Questions (40 points)

Answer questions 1-3 according to the following graph.



1. In the graph above, what is the maximum flow from s to t?

a) 3
b) 4
c) 5
d) 6
e) 7

2. In the graph above, what is the minimum cut value from s to t?

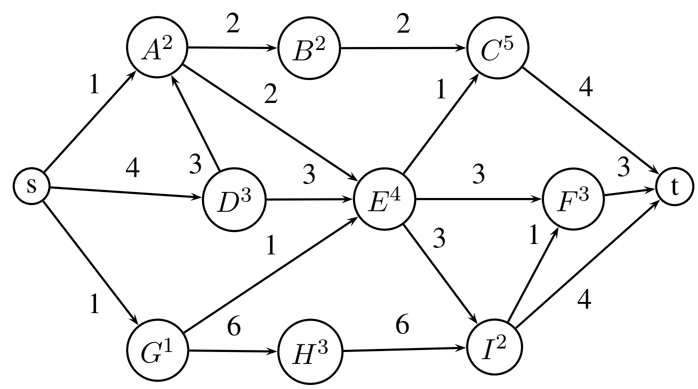
a) 3
b) 4
c) 5
d) 6
e) 7

3. In the graph above, what is the minimum cut from s to t (S-component, T-component)?

a) (S, A-B-C-D-E-F-G-H-I-t)
b) (S-A-G, B-C-D-E-F-H-I-t)
c) (S-A-B-C-D-E-F-G-H-I, t)
d) (S-A-D-G, B-C-E-F-G-H-I, t)
e) None of the above

4. How can we convert multi-source / multi-sink problem to single-source / single-sink problem?

a) Connect each source directly to each sink.
b) Create a super source, connect super source to all sources, then connect each source directly to each sink.
c) Create a super sink, connect each source to this super sink, then connect each sink to this super sink.
d) Create a super source and super sink, connect super source to all sources, then connect all sinks to the super sink.
e) Any one of the above



5. Given the vertex capacities above the letters, what is the maximum flow from s to t?

a) 3
b) 4
c) 5
d) 6
e) 7

6. When a network contains also vertex capacities, we use vertex splitting technique on the graph to solve the problem. Let say in the beginning there are V vertices in the graph, what will be the number of vertices in the resulting graph after vertex splitting?

a) V - 1
b) V / 2
c) V + 1
d) 2V
e) V^2

7. If we try to construct all possible subsets of an n element set via backtracking, how many candidates will be generated per construct_candidates method call?

a) n
b) 1
c) 2^n
d) 3
e) 2

8. If we try to construct all numbers with n digits via backtracking, how many candidates will be generated per construct_candidates method call?

a) n
b) 10
c) 10^n
d) 2
e) 9

9. If we try to construct all possible permutations of an n element set via backtracking, how many candidates will be generated per construct_candidates method call for the element with index k?

a) n
b) k
c) $n!$
d) $k!$
e) $n - k$

10. Let say we are constructing all possible subsets of an 4 element set via backtracking, and let also say that we are at the candidate FFTF, what will be the next candidate?

a) TTTF
b) FFTT
c) FTTF
d) TFTF
e) FFFF

11. Let say we are constructing all numbers with 4 digits via backtracking, and let also say that we are at the candidate 2345, what will be the next candidate?

a) 2335
b) 1345
c) 1445
d) 2346
e) 2342

12. Let say we are constructing all strings with 4 characters via backtracking, and let also say that we are at the candidate ‘cefg’, what will be the next candidate?

a) defg
b) cffg
c) cefh
d) cefi
e) cegg

13. Is P = NP?

a) Yes
b) No
c) Unknown
d) Can not be proven
e) None of the above

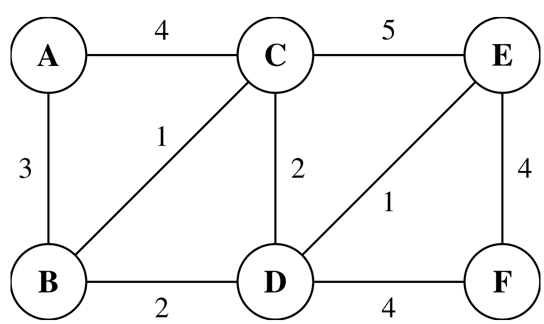
14. Which of the following is an example P problem?

a) 2SAT
b) Minimum Spanning Tree
c) Shortest Path
d) Eulerian Path
e) All of the above

15. Which of the following is an example NP-complete problem?

a) 3SAT
b) Traveling Salesman Problem
c) Rudrata Path
d) Integer Linear Programming
e) All of the above

16. Given the graph below, what is the longest path



from A to F?

a) 13
b) 14
c) 15
d) 16
e) 17

17. What is the time complexity of the brute force search of the Traveling Salesman Problem with N cities?

a) 3^N
b) 2^N
c) N!
d) $\binom{N}{N/2}$
e) None of the above

18. What is the time complexity of the brute force search of the 3-SAT Problem with N variables?

a) 3^N
b) 2^N
c) N!
d) $\binom{N}{N/2}$
e) None of the above

19. What is the time complexity of the brute force search of the Rudrata Cycle Problem with N vertices?

a) 3^N
b) 2^N
c) N!
d) $\binom{N}{N/2}$
e) None of the above

20. What is the time complexity of the brute force search of the ZOE Problem with N variables?

a) 3^N
b) 2^N
c) N!
d) $\binom{N}{N/2}$
e) None of the above